

Logic Programs with Negation

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Syntax of Normal LPs

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Syntax of Normal LPs

Suppose $A, B_1, \dots, B_m, D_1, \dots, D_n$ are atoms.

Normal Rule

$$A \leftarrow B_1, \dots, B_m, \neg D_1, \dots, \neg D_n$$

is a **normal rule**. A **normal logic program** is a finite set of normal rules.

Examples

P_1 :

$$p \leftarrow \neg q.$$
$$q \leftarrow r.$$

P_2 :

$$p \leftarrow \neg q.$$
$$q \leftarrow r \wedge s.$$
$$r \leftarrow s.$$

Semantics of Normal LPs

Semantics of Normal LPs

Intuition

- ① We want to interpret the $\neg$ symbol as “not proved”.
- ② E.g. Suppose the rule $p \leftarrow \neg q$ is present in some logic program P . The meaning LPers wanted to ascribe to this rule was “If q is not provable from P then infer p .”
- ③ Is there a problem?

Stable Semantics: Basic Idea

- ① Guess an interpretation I consisting of a set of all ground atoms that are possibly provable from P .
- ② Use I to interpret $\neg$ as not “possibly” provable.
- ③ If I is correct, we can eliminate some rules from P and also eliminate negations from the remaining rules.
- ④ The result is a definite (negation free) logic program $G(P, I)$.
- ⑤ The set of atoms necessarily provable from P if I is assumed to be what is provable is $T_{G(P, I)} \uparrow \omega$.
- ⑥ We guessed well if everything that if the set of necessarily provable atoms coincide with our assumption that I is the set of possibly provable atoms, i.e. $I = T_{G(P, I)} \uparrow \omega$. In this case, we say that I is stable.

Examples

P_1 :

$$p \leftarrow \neg q.$$

$$q \leftarrow r.$$

Suppose $I = \{\}$. Then $G(P, I)$ is the set:

$$p \leftarrow .$$

$$q \leftarrow r.$$

$T_{G(P, I)} \uparrow \omega = \{p\}$, so $I = \{\}$ is not stable.

Examples

P_1 :

$$p \leftarrow \neg q.$$

$$q \leftarrow r.$$

Suppose $I = \{p\}$. Then $G(P, I)$ is the set:

$$p \leftarrow .$$

$$q \leftarrow r.$$

$T_{G(P, I)} \uparrow \omega = \{p\}$, so $I = \{p\}$ is stable.

Example

P_2 :

$$p \leftarrow \neg q.$$
$$q \leftarrow r \wedge s.$$
$$r \leftarrow s.$$

Again, the only stable model is $I = \{p\}$.

Theorem

Theorem

Suppose P is a normal logic program and I is stable. Then I is a model of P .

Proof sketch.

- Suppose I is stable and $A \leftarrow B_1, \dots, B_m, \neg D_1, \dots, \neg D_n$ is a ground instance of some rule in P .
- Suppose $I \models B_1, \dots, B_m, \neg D_1, \dots, \neg D_n$ — we need to show that $A \in I$.
- If $I \models B_1, \dots, B_m, \neg D_1, \dots, \neg D_n$, then for all $1 \leq i \leq n$, $D_i \notin I$. So $A \leftarrow B_1, \dots, B_m \in G(P, I)$.
- But as I is stable, $I = T_{G(P, I)} \uparrow \omega$, so I is a model of $G(P, I)$. As $\{B_1, \dots, B_m\} \subseteq I$, it follows that $A \in I$.

Some Questions for Discussion

Suppose you are given a normal logic program P .

- 1 Is P guaranteed to have a stable model? **NO**.
- 2 If a stable model exists, is it guaranteed to be unique? **NO**.

Theorem

Theorem

If I is stable w.r.t. logic program P , then I is a model of P .

Proof.

- Suppose I is stable w.r.t. P and suppose $A \leftarrow B_1, \dots, B_m, \neg D_1, \dots, \neg D_n$ is a ground instance of a rule in P such that $I \models B_1 \wedge \dots \wedge B_m \wedge \neg D_1 \wedge \dots \wedge \neg D_n$, i.e. $\{B_1, \dots, B_m\} \subseteq I$ and $\{D_1, \dots, D_n\} \cap I = \emptyset$.
- As $\{D_1, \dots, D_n\} \cap I = \emptyset$, $A \leftarrow B_1, \dots, B_m$ is in $G(P, I)$.
- As $\{B_1, \dots, B_m\} \subseteq I$, $A \in T_{G(P, I)}(I)$.
- But $I = \text{lfp}(T_{G(P, I)})$ because I is stable, so it is a fixpoint of $T_{G(P, I)}$. Hence $A \in I$.

F_P Operator

F_P Operator

Suppose P is a normal logic program and I is an interpretation.

Then: $F_P(I) = T_{G(P,I)} \uparrow \omega = \text{Ifp}(T_{G(P,I)})$.

$G(P, I)$ is Anti-Monotonic

Proposition

Suppose P is a normal LP and $I_1 \subseteq I_2$ are two Herbrand interpretations. Then: $G(P, I_2) \subseteq G(P, I_1)$.

Proof. Suppose $I_1 \subseteq I_2$ and $A \leftarrow B_1, \dots, B_m$ is in $G(P, I_2)$.

- Then there is a rule of the form $A \leftarrow B_1, \dots, B_m, \neg D_1, \dots, \neg D_n$ in P such that $\{D_1, \dots, D_n\} \cap I_2 = \emptyset$.
- As $I_1 \subseteq I_2$, it follows that $\{D_1, \dots, D_n\} \cap I_1 = \emptyset$.
- But then $A \leftarrow B_1, \dots, B_m$ is in $G(P, I_1)$.

F_P is Anti-Monotonic

Proposition

Suppose P is a normal LP and $I_1 \subseteq I_2$ are two Herbrand interpretations. Then: $F_P(I_2) \subseteq F_P(I_1)$.

Proof. Suppose $I_1 \subseteq I_2$ and $A \in F_P(I_2)$.

- Then $A \in T_{G(P, I_2)} \uparrow \omega$ and hence there is some integer k s.t. $A \in T_{G(P, I_2)} \uparrow k$. We proceed by induction on k .
- **Base Case ($k = 1$)** Then $A \leftarrow$ is a ground instance of a rule in $G(P, I_2)$. As $G(P, I_2) \subseteq G(P, I_1)$ by the previous theorem, $A \leftarrow$ is a g.i. of a rule in $G(P, I_1)$ and hence is in $T_{G(P, I_1)} \uparrow 1 \subseteq F_P(I_1)$.
- **Inductive Case ($k = r + 1$)** Next slide.

F_P is Anti-Monotonic (Proof continued)

- If $A \in T_{G(P, l_2)} \uparrow (r + 1)$, there is a rule of the form $A \leftarrow B_1, \dots, B_m$ in $G(P, l_2)$ s.t. $\{B_1, \dots, B_m\} \subseteq T_{G(P, l_2)} \uparrow r$.
- By the IH, $T_{G(P, l_2)} \uparrow r \subseteq F_P(G, l_1) = T_{G(P, l_1)} \uparrow \omega$.
- There must exist an integer s s.t. $\{B_1, \dots, B_m\} \subseteq T_{G(P, l_1)} \uparrow s$.
- By the previous theorem, the rule $A \leftarrow B_1, \dots, B_m$ in $G(P, l_1)$.
- From the previous two bullets, $A \in T_{G(P, l_1)} \uparrow (s + 1) \subseteq T_{G(P, l_1)} \uparrow \omega = F_P(l_1)$.

F_P^2 is Monotonic

Corollaries

- 1 F_P^2 is monotonic and has a least fixpoint $lfp(F_P^2)$ and a greatest fixpoint $gfp(F_P^2)$.
- 2 If I is a stable model of P , then $lfp(F_P^2) \subseteq I \subseteq GFP(F_P^2)$.

Iterations of F_P

$$\begin{aligned} F_P^0 &= \emptyset. \\ F_P^{i+1} &= F_P(F_P^i). \end{aligned}$$

Theorem

Theorem

- ① For all *odd* integers i , $\text{gfp}(F_P) \subseteq F_P^i$.
- ② For all *odd* integers i , $F_P^i \supseteq F_P^{i+2}$.
- ③ For all *even* integers i , $\text{lfp}(F_P) \supseteq F_P^i$.
- ④ For all *even* integers i , $F_P^i \subseteq F_P^{i+2}$.
- ⑤ $F_P(\text{lfp}(F_P^2)) = \text{gfp}(F_P^2)$.
- ⑥ $F_P(\text{gfp}(F_P^2)) = \text{lfp}(F_P^2)$.

Proofs will be done in class on the board.